Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes

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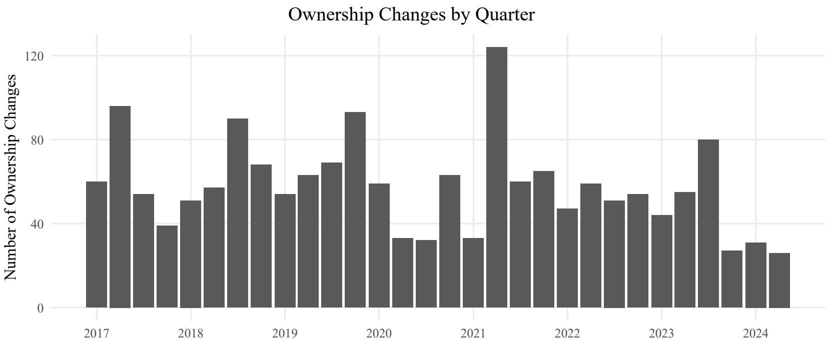
Motivation: Ownership Changes Beyond Market Structure

- Ownership changes are very common
  - (1) Mergers and Acquisitions
  - (2) New Management
- This second type of changes does **not** change the set of providers in a market.
- Instead, effects operate through **internal production decisions**
- This second type is understudied in the literature and we study this issue in **nursing homes**

Ownership Changes Over Time (Quarterly)



Why Nursing Homes?

- There are about 15,000 nursing homes in the U.S. that serve over 4 million people annually
- 56% of adults aged 57 to 61 will spend at least one night in a nursing home during their lifetime
- Nursing home care is labor-intensive and heavily regulated.
- **Nursing labor is the main input** into care and a major operating cost.
- Staffing is a natural adjustment margin after ownership change, but it's constrained by:
  - compliance,
  - resident outcomes,
  - demand/occupancy

Staffing is Important

- Staffing is the primary input into nursing home care, so ownership changes can affect quality even when market structure is unchanged.
- Higher staffing levels are linked to better quality measures in nursing homes, including fewer deficiency citations (Harrington et al., 2000).
- Causal evidence shows that increasing nurse staffing improves measured nursing home quality (Lin, 2014).
- If ownership changes shift staffing decisions, they can translate into meaningful differences in care quality.

Research Question

- **Research question:** How do nurse staffing levels respond to ownership changes in nursing homes?
    - Do responses differ across staff type?
    - Do responses differ if a nursing home is part of a chain or not?
    - Do responses differ pre vs post-pandemic?

Nursing Staff Types and Staffing Measurement

- **Nursing homes staff care using three main nursing roles:**
  - **RN (Registered Nurses):** clinical assessment, care planning, meds, supervision
  - **LPN (Licensed Practical Nurses):** routine clinical tasks, meds under RN oversight
  - **CNA (Certified Nursing Assistants):** direct daily care
  - **Total nursing:** RN + LPN + CNA
- **Staffing measure: Hours Per Patient Day (HPPD)**
  - Standardizes staffing by resident census (staffing intensity, not raw hours)
  - Lets us compare facilities and months on a common scale
- Ownership change can affect both **levels** (total HPPD) and **mix** (RN/LPN/CNA composition) which sets up cost-quality tradeoff.

The Cost–Quality Tradeoff in Staffing

- Owners can raise quality through two channels:
  - **Quantity (levels):** increasing total nursing hours (HPPD)
  - **Skill mix:** shifting hours toward more clinically trained staff (e.g., more RN oversight)
- **Both margins raise costs** because nursing labor is a major operating expense.
- **But short-run revenues are relatively fixed** (public payers, limited pricing flexibility), so staffing choices map into a **cost–quality tradeoff**.
- **Implication:** an ownership change that shifts objectives or constraints can move a facility along this tradeoff. (Park and Werner, 2011; Bowblis and Brunt, 2025)

Why Staffing Could Rise or Fall After a Transition

- Ownership changes can shift **constraints** and **objectives** $\Rightarrow$ staffing effects are ambiguous.
- **Scenario 1: Cost pressure $\Rightarrow$ staffing falls**
  - Debt service / tighter targets $\Rightarrow$ cut labor costs (lower HPPD or cheaper mix)
- **Scenario 2: Quality/compliance push $\Rightarrow$ staffing rises**
  - Invest in staffing to improve oversight / survey performance (higher HPPD or better mix)
- **Scenario 3: Reorganization $\Rightarrow$ mix changes**
  - Standardized staffing model shifts RN/LPN/CNA composition; total hours may be similar

Data Sources and Unit of Observation

- **Healthcare Cost Report Information System (HCRIS)**
  - Annual cost report information
  - Contains information about costs, revenue, financial data and dates of ownership change
- **Nursing Home Compare (NHC) Archive**
  - Contains ownership information as well as facility characteristics
- **Payroll Based Journal (PBJ)**
  - Payroll-reported nurse staffing hours by staff type
  - Used to construct outcome variables
- **Facility-month panel, 48 contiguous states, Jan 2017 – Jun 2024**

Identifying Ownership Changes

- We identify ownership changes using a 3 step process
- **Step 1 (HCRIS):** includes a variable indicating **whether and when** an ownership change occurred.
- **Challenge:** ownership is often **layered** (organizations owning organizations), so an administrative flag may not map cleanly to an **economic change in control**.
- HCRIS might not contain all ownership changes, or it may contain those with no meaningful economic change, so we have to verify using the NHC Archive data.

Ownership Level Challenge

- We identify ownership changes at the **highest level of ownership present**

Ownership Level	Type	Owner Name	Share
Direct ownership			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
Organization	LLC	ROCK HILL ACQUISITION III LLC	75%
Indirect ownership			
Individual	Person	LALLY, THOMAS	25%
Individual	Person	LITT, ALAN	25%
Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

Identifying Ownership Changes

- **Step 2: Validate & date using NHC**
  - Track ownership shares over time within the highest reported ownership level
  - Define a change when **majority control (50%)** transfers between $t-1$ and t
  - We do not flag changes driven by name changes or within-family reclassification
- **Example from the data:**
  - *Before:*
    - ESTES JAMES (84%)
    - BURCHFIELD JOHN (10%)
    - LEE CLAUDE (6%)
  - *After:*
    - JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
    - JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)
  - We do **not** treat this as a meaningful ownership change

Ownership Change Agreement: NHC vs HCRIS

- **Validate HCRIS ownership-change flags using NHC ownership records.**
- We keep facilities only if:
  - **Never-treated:** 0 changes in **both** NHC and HCRIS, or
  - **Single change:** exactly 1 change in **each** source, within **6 months**.
- **Event month:** first month NHC shows a majority-control transfer.

Ownership Change Agreement: NHC vs HCRIS (Cross-tab)

Cross-tab of Ownership Changes

Count of changes in NHC \ Count of changes in HCRIS	0	1	2+
2+	1,137 (8.5%)	1,377 (10.3%)	535 (4.0%)
1	1,874 (14.0%)	1,589 (11.9%)	101 (0.8%)
0	6,217 (46.6%)	499 (3.7%)	15 (0.1%)

Dependent Variables: Nurse Staffing (PBJ) and Analysis Sample

- We study four staffing measures:
  - RN, LPN, CNA, and total HPPD
- Outcome is **hours per patient day (HPPD)**: staffing intensity standardized by resident census

$$\text{HPPD}_{s,i,t} = \frac{H_{s,i,t}}{\text{PatientDays}_{i,t}}$$

- **CMS implausible staffing exclusions:**
  - RN HPPD $= 0$ **and** LPN HPPD $= 0$
  - Total HPPD < 1.5 or > 12
  - CNA HPPD > 5.25
- **Additional restrictions:** beds < 15; hospital-based facilities
- **Final sample:** 7,806 facilities; 632,231 facility-months; 1,589 treated ($\approx 20\%$)

Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD around ownership transitions Goodman-Bacon, 2021
- **Variation:** within-facility staffing changes relative to facilities *not yet treated*
- **Event study regression:**

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_\tau \, \mathbf{1}\{event\_time_{it} = \tau\} \cdot Treat_i + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}.$$

- **Key objects:**
  - Y_{it}: staffing outcome in **HPPD** (RN, LPN, CNA, Total)
  - β_τ: effect at event time τ (relative to the reference period)
  - α_i facility FE; λ_t month FE; X_{it} controls (beds, occupancy, payer mix, ownership type, chain, acuity)
  - SEs clustered at the facility level

Empirical Strategy: TWFE Difference-in-Differences

- **Goal:** summarize the **average** staffing change after an ownership transition
- **TWFE DiD regression:**

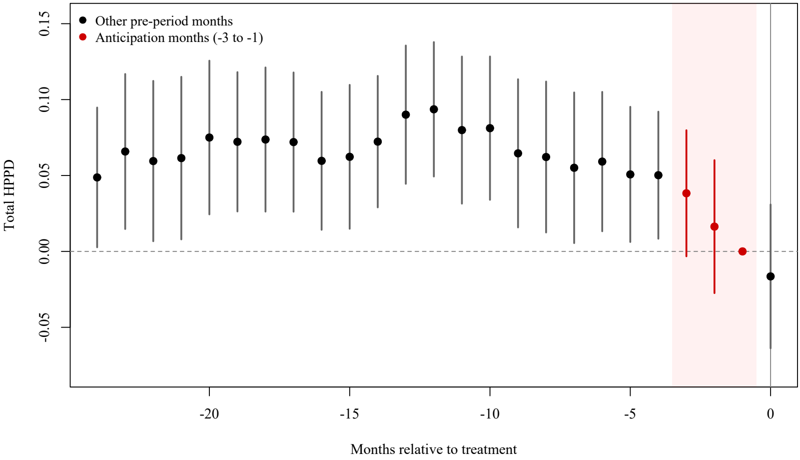
$$Y_{it} = \beta\, Post_{it} + X_{it}'\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- **Key objects:**
  - $Post_{it} = 1$ in all months after a change ($= 0$ before)
  - β: **average post-change effect** on staffing (HPPD)
  - Same fixed effects and controls as the event-study specification
  - SEs clustered at the facility level

Identification: Parallel Trends

- **Parallel trends:** absent an ownership change, treated facilities would have followed the same staffing trends as facilities not yet treated.
- **Staffing may change in anticipation of a formal ownership change**
  - Changes are not instantaneous: negotiations and operational handoffs precede the recorded month
  - Staffing can adjust during planning (turnover, hiring freezes, renegotiated contracts)
  - Recorded dates may reflect administrative/legal timing, not first operational change
- **Implication:** months just before $\tau = 0$ may violate parallel trends.

Evidence of violation in the Pre-Period



Main Specification: Donut Event Study + Pre-trend Test

- **Donut event study:** mitigate this influence by excluding months just before the recorded transition.

- **Implementation**
  - Exclude $\tau \in \{-3, -2, -1\}$
  - Use $\tau = -4$ as the reference period

- **Parallel Trends:** joint Wald test of pre-period leads

$$H_0 : \ \beta_\tau = 0 \quad \forall \, \tau < 0$$

- **Takeaway:** full pre-window shows evidence of pre-trends; the donut specification is more consistent with parallel trends.

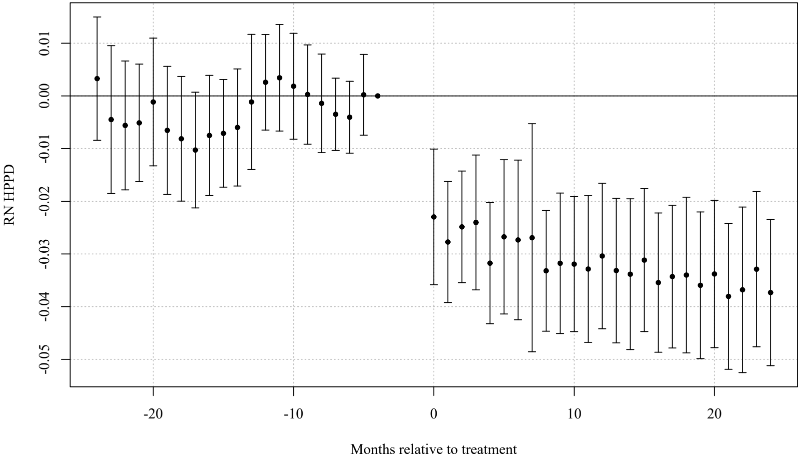
Parallel Trends Tests (Joint Wald) — Total Staffing

Table: **Pre-trend Wald Test p-values (Total Staffing)**

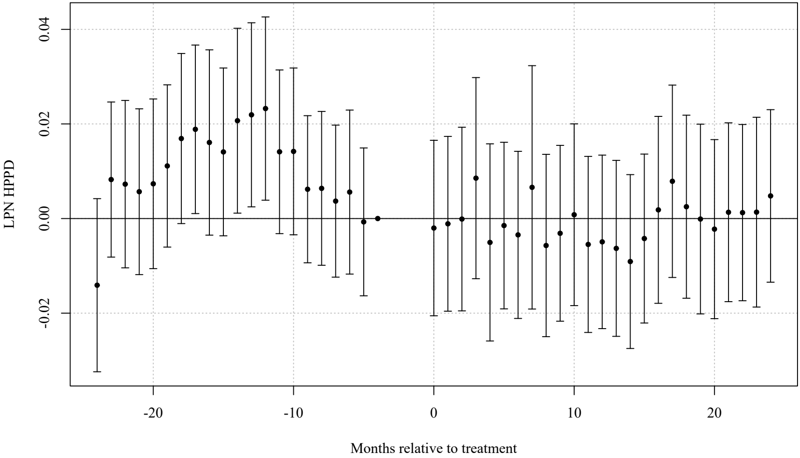
	Total (Levels)	**Total (Logs)**
2 Year Full Pre-Window	0.0006	0.0012
2 Year Window with Donut	0.2046	0.3268
1 Year Window with Donut	0.6387	0.6102

Notes: Joint Wald χ^2 test of $H_0 : \beta_\tau = 0$ for all pre-treatment coefficients. Cells report p-value.
Donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.
Full outcome-by-staff-type table available in appendix.

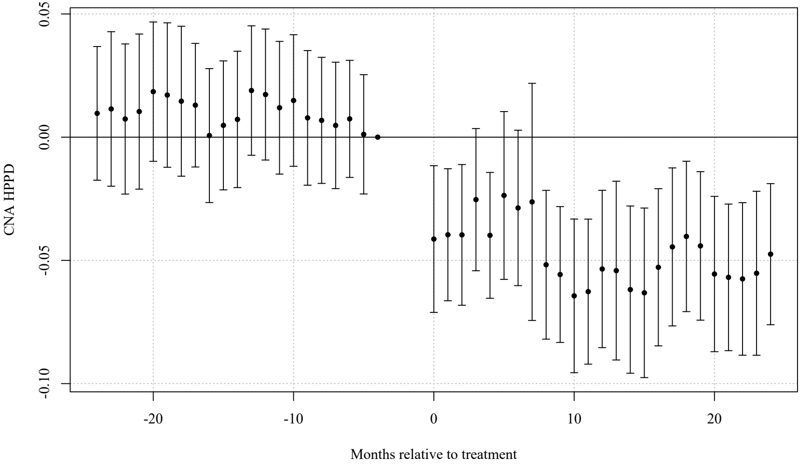
Event Study Results (RN HPRD)



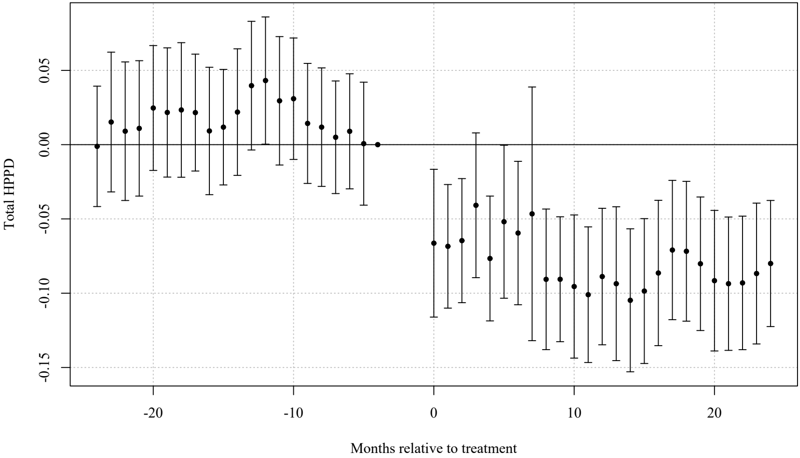
Event Study Results (LPN HPRD)



Event Study Results (CNA HPRD)



Event Study Results (Total Nursing HPRD)



TWFE DiD — Baseline (RN)

	RN	LPN	CNA	Total
HPRD	-0.032^{***}	0.000	-0.054^{***}	-0.086^{***}
	(0.005)	(0.006)	(0.009)	(0.013)
Log(HPRD)	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
	(0.016)	(0.009)	(0.005)	(0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. $N = 628{,}107$

TWFE DiD — Baseline (LPN)

	RN	LPN	CNA	Total
HPRD	-0.032^{***}	0.000	-0.054^{***}	-0.086^{***}
	(0.005)	(0.006)	(0.009)	(0.013)
Log(HPRD)	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
	(0.016)	(0.009)	(0.005)	(0.004)

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TWFE DiD — Baseline (CNA)

	RN	LPN	CNA	Total
HPRD	-0.032^{***}	0.000	-0.054^{***}	-0.086^{***}
	(0.005)	(0.006)	(0.009)	(0.013)
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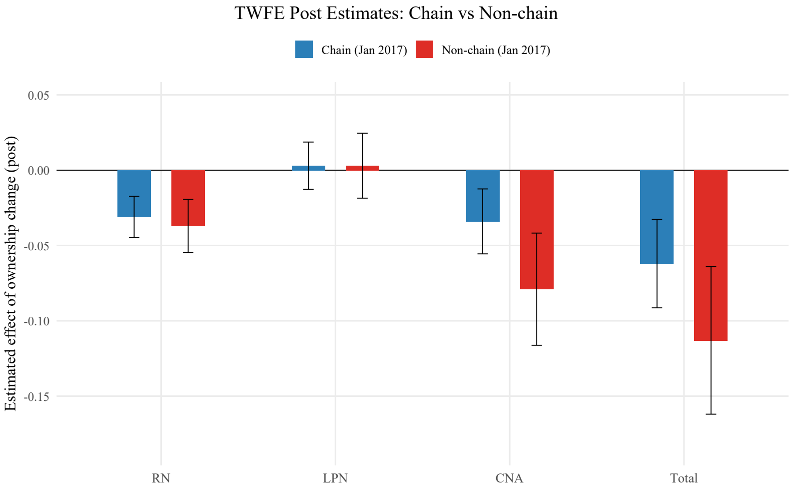
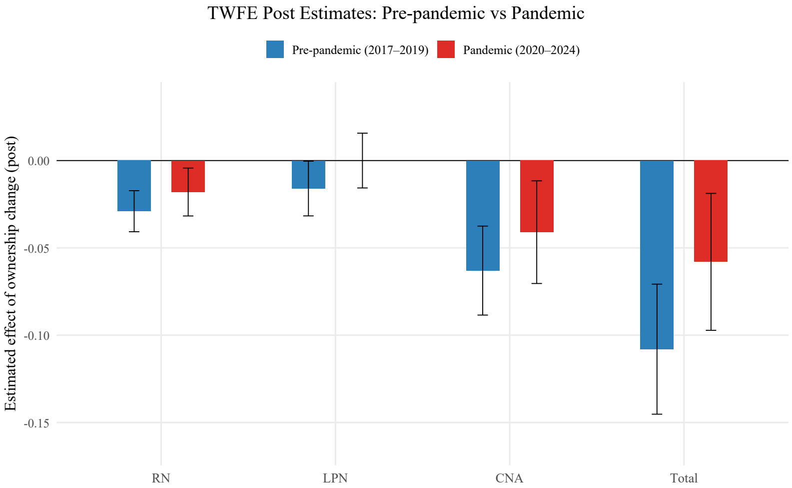
Heterogeneity: Chain vs Non-Chain Facilities



Staffing declines are larger for non-chain facilities, nearly double for CNA and Total.

Heterogeneity: Pre vs Post-Pandemic



Large pre-pandemic declines, and post-pandemic effects are almost cut in half.

Robustness Checks

- **Control-group**
  - Re-estimate using **not-yet-treated** controls only.
  - Re-estimate using **never-treated** facilities only.
  - **Stacked DiD** (cohort-by-cohort).
- **Treatment timing / definition.**
  - Shift event date by ± 1 month.
  - NHC vs HCRIS

Robustness Checks

- **Donut Hole Specification**
  - Baseline uses a **donut**: drop $\tau \in \{-3, -2, -1\}$ (reference $\tau = -4$).
  - **Wider donut:** drop $\tau \in \{-4, -3, -2, -1\}$.
  - **Smaller donut:** drop only $\tau \in \{-2, -1\}$.
  - Estimates are unchanged in sign, magnitude, and significance.

- **Reporting gaps in staffing data.**
  - Restrict sample to facilities with smaller gaps between observations (exclude gaps > 6, > 3, > 1, and > 0 months).
  - Estimates are nearly identical to baseline.

Discussion: Why do LPN hours not decline?

- Roles differ by scope-of-practice:
  - **RN:** clinical oversight & supervision
  - **CNA:** hands-on ADLs / direct care
  - **LPN:** middle-skill routine clinical + direct care overlap
- **Substitution:** LPNs can cover some RN *and* CNA tasks ⇒ "buffer" when RN/CNA hours fall
- **Implication:** ownership changes may shift **skill mix** more than cut all staffing uniformly

(e.g., Lin 2014; Harrington et al. 2000; Bowblis & Ghattas 2017)

Magnitude: statistically significant vs meaningful

- Baseline, minutes per resident-day:
  - **RN:** -0.032 HPPD ≈ -1.9 min
  - **CNA:** -0.054 HPPD ≈ -3.2 min
  - **Total:** -0.086 HPPD ≈ -5.2 min
- Benchmarks (sample means): **RN $\sim$7–9%**, **Total $\sim$2–3%** (logs similar)
- **Key point:** big samples $\Rightarrow$ precise estimates; significance $\neq$ clinical impact

When might it matter? Facility size & diffusion

- Same average change can play out differently across facilities
- Example: Total -0.086 HPPD $\Rightarrow$ 8.6 staff-hours/day at 100 residents
- **Medium/large:** spread across shifts/residents $\Rightarrow$ likely absorbed
- **Small:** More likely to affect quality of care

Heterogeneity: Chains and Covid

- **Non-chain facilities:** larger declines (fewer organizational buffers)
- **COVID period:** smaller declines (labor market tightness / oversight constraints)
- Suggests effects depend on **organizational capacity** and **labor market conditions**

Limitations and future extensions

- **Timing noise / anticipation:** motivates donut window
- **Remaining confounding:** ownership may coincide with distress/restructuring
- **Next step:** link to resident outcomes / quality metrics (deficiencies, rehospitalizations, mortality, stars)
- Distinguish: **efficiency/task reallocation** vs **quality-reducing cost cutting**

Conclusion

- Ownership changes lead to persistent declines in **RN/CNA/Total** staffing; **LPN stable**
- Average magnitudes modest (minutes per resident-day; small % for total)
- Effects larger for **non-chains** and smaller during **COVID**
- Takeaway: ownership changes affect care via **internal production decisions**, even without market-structure change

Thank you

Questions?

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